

ASHLEE FAROTTE-KRUCHAS

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PROFESSIONAL SUMMARY

Physics and biology student with a mathematics minor and hands-on research experience in computational astronomy, machine learning, scientific data analysis, and vertebrate paleontology preparation. Experienced with Python-based research tools, large scientific datasets, laboratory procedures, and high-performance computing. Detail-oriented, adaptable, and eager to apply a multidisciplinary science background while learning new laboratory and research techniques.

EDUCATION

University of Oklahoma — Norman, OK

B.S. candidate, Biology and Physics (dual major), expected Spring 2027

Minor in Mathematics — requirements completed

GPA: 3.5+

RESEARCH EXPERIENCE

Undergraduate Researcher | University of Oklahoma

Norman, OK | 2026 – Present

- Conduct computational astronomy research involving stellar spectroscopy, machine learning, and large scientific datasets.
- Use Python to process, analyze, organize, and compare large scientific datasets.
- Develop and train neural-network models to study relationships between stellar properties and observed spectra.
- Combine and crossmatch astronomical catalogs to build validation datasets for research models.
- Retrieve, prepare, and validate observational data for scientific analysis and model evaluation.
- Use high-performance computing resources and scientific Python tools to manage data, run analyses, and train models.
- Work under research mentorship to troubleshoot code, evaluate results, document methods, and learn new technical tools and research methods.

SCIENCE & LABORATORY EXPERIENCE

Vertebrate Paleontology Preparation Lab Volunteer | Sam Noble Museum

Norman, OK | 11/2023 – 12/2024

- Carefully unpackaged, cleaned, sorted, and prepared prehistoric macro- and microvertebrate fossils for scientific research and observation.
- Created molds to reconstruct incomplete or damaged specimens and followed detailed preparation instructions from laboratory supervisors.
- Used handheld and electromechanical tools and laboratory chemicals, including dental picks, air scribes, cast-cutter saws, acetone, and bonding materials.

School Aquarium Program Volunteer | Cabrillo High School

Lompoc, CA | 08/2019 – 04/2020

- Provided daily animal care, monitored aquarium animals for unusual medical or behavioral signs, and reported concerns to the aquarium manager.
- Led educational tours and interactive science activities for elementary students, senior citizens, special-needs students, and the general public.
- Maintained tanks, glass, animal enclosures, and program spaces.

ADDITIONAL WORK EXPERIENCE

Server | Olive Garden

Midwest City, OK | 02/2023 – 03/2024

- Managed multiple guest needs in a fast-paced environment while maintaining accuracy, organization, teamwork, and customer service.

Shift Lead Barista / Trainer | Scooter's Coffee

Midwest City, OK | 05/2022 – 02/2023

- Trained new employees, supervised shift operations, tracked sales and speed metrics, handled bank runs and truck orders, and supported employee development.
- Maintained quality, efficiency, cash-handling accuracy, equipment cleanliness, and health-regulation compliance.

TECHNICAL & RESEARCH SKILLS

Computational: Python, NumPy, pandas, TensorFlow/Keras, h5py, Astropy, Matplotlib, SLURM, high-performance computing

Research: Scientific data analysis, machine learning, stellar spectroscopy, catalog crossmatching, research documentation, troubleshooting

Laboratory: Specimen preparation, careful tool use, chemical handling, following laboratory protocols, animal observation and care

HONORS & ACTIVITIES

Honor Society • OU Botany Club • OU Bee Club • OU CKI